

Fraction Universe

Every graded set, every week — 6 weeks, 30 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

WEEK 6

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

One Whole, Cut Up

Answers are whole numbers — when a question asks for a numerator, type just that number.

Warm-Up 6 ITEMS

Halves in one whole

Quarters in one whole

Eighths in one whole

$1/2 = ?/4 \rightarrow$ numerator

$1/2 = ?/8 \rightarrow$ numerator

$3/3 =$ how many wholes?

Core GRADED • SHOW YOUR WORKING

1. $1/4 + 1/4 + 1/4 = ?/4 \rightarrow$ numerator (Count the quarters)

.....

2. How many thirds in 2 wholes? (Three in each whole)

.....

3. $5/5 =$ how many wholes? (All five pieces)

.....

4. $7/4$ — how many whole ones? ($4/4$ makes one whole)

.....

5. $7/4$ — how many quarters left over? ($7 - 4$)

.....

Challenge NOT GRADED

C1. Marks between 0 and 1 when you cut into fifths (not counting 0 and 1) (Five pieces need four cuts)

.....

C2. $3/8$ and $5/8$ — how many eighths apart? ($5 - 3$)

.....

Same Number, New Name

Multiply top and bottom by the same thing and nothing about the amount changes.

Warm-Up 6 ITEMS

$1/2 = ?/6 \rightarrow$ numerator

$1/3 = ?/6 \rightarrow$ numerator

$2/3 = ?/6 \rightarrow$ numerator

$1/4 = ?/8 \rightarrow$ numerator

$3/4 = ?/8 \rightarrow$ numerator

$1/5 = ?/10 \rightarrow$ numerator

Core GRADED • SHOW YOUR WORKING

1. $3/4 = ?/12 \rightarrow$ numerator (The bottom was tripled)

2. $2/5 = ?/20 \rightarrow$ numerator (The bottom was multiplied by 4)

3. $6/8$ in quarters = $?/4 \rightarrow$ numerator (Divide both by 2)

4. $10/15$ simplified $\rightarrow$ numerator (Divide both by 5)

5. $9/12$ simplified $\rightarrow$ denominator (Divide both by 3)

Challenge NOT GRADED

C1. $8/12 = ?/3 \rightarrow$ numerator (Divide both by 4)

C2. How many fractions equal $1/2$ with a denominator under 20? (2, 4, 6 ... 18)

Which Is Bigger

Make the pieces the same size first. Then it's just counting.

Warm-Up 6 ITEMS

Bigger: $1/2$ or $1/4$ → its denominator

Bigger: $1/3$ or $1/5$ → its denominator

Bigger: $3/4$ or $1/4$ → its numerator

Bigger: $2/5$ or $3/5$ → its numerator

Bigger: $5/6$ or $5/8$ → its denominator

Bigger: $7/10$ or $3/10$ → its numerator

Core GRADED • SHOW YOUR WORKING

1. $2/3$ or $3/5$ in fifteenths → bigger numerator (10/15 against 9/15)

2. $3/4$ or $5/8$ in eighths → bigger numerator (6/8 against 5/8)

3. $5/6$ or $7/9$ in eighteenths → bigger numerator (15/18 against 14/18)

4. Closer to 1: $7/8$ or $5/6$ → its denominator (A gap of $1/8$ beats a gap of $1/6$)

5. $2/3$ and $3/4$ in twelfths → bigger numerator (8/12 against 9/12)

Challenge NOT GRADED

C1. Largest of $3/5$, $2/3$, $7/10$ in thirtieths → its numerator (18/30, 20/30, 21/30)

C2. Halfway between $1/2$ and $3/4$ as $?/8$ → numerator (Between 4/8 and 6/8)

Adding Needs Matching Pieces

Change one fraction, or both, until the bottoms agree. Then count.

Warm-Up 6 ITEMS

$$1/4 + 1/4 = ?/4 \rightarrow \text{numerator}$$

$$2/5 + 1/5 = ?/5 \rightarrow \text{numerator}$$

$$3/8 + 2/8 = ?/8 \rightarrow \text{numerator}$$

$$5/6 - 2/6 = ?/6 \rightarrow \text{numerator}$$

$$7/10 - 4/10 = ?/10 \rightarrow \text{numerator}$$

$$3/4 + 1/4 = \text{how many wholes?}$$

Core GRADED • SHOW YOUR WORKING

1. $1/2 + 1/4 = ?/4 \rightarrow \text{numerator } (2/4 + 1/4)$

2. $2/3 + 1/6 = ?/6 \rightarrow \text{numerator } (4/6 + 1/6)$

3. $3/4 - 1/8 = ?/8 \rightarrow \text{numerator } (6/8 - 1/8)$

4. $5/6 - 1/3 = ?/6 \rightarrow \text{numerator } (5/6 - 2/6)$

5. $1/2 + 1/3 = ?/6 \rightarrow \text{numerator } (3/6 + 2/6)$

Challenge NOT GRADED

C1. $3/4 + 2/3 = ?/12 \rightarrow \text{numerator } (9/12 + 8/12)$

C2. $2 - 3/5 = ?/5 \rightarrow \text{numerator } (10/5 - 3/5)$

Fraction of a Set & Closest to One

Enrichment day. Divide by the bottom, multiply by the top — always that order.

★ Challenge NOT GRADED

C1. $\frac{1}{4}$ of 20 ($20 \div 4$)

C2. $\frac{2}{5}$ of 30 ($30 \div 5 = 6$, then $\times 2$)

C3. $\frac{3}{8}$ of 40 ($40 \div 8 = 5$, then $\times 3$)

C4. $\frac{2}{3}$ of 45 ($45 \div 3 = 15$, then $\times 2$)

C5. $\frac{5}{6}$ of 42 ($42 \div 6 = 7$, then $\times 5$)

C6. $\frac{3}{4}$ of a number is 18. The number? ($18 \div 3 = 6$ is one quarter)

C7. Closest to 1 without going over: $\frac{5}{6}$, $\frac{7}{8}$, $\frac{9}{10}$ → its denominator (Gaps of $\frac{1}{6}$, $\frac{1}{8}$, $\frac{1}{10}$)

C8. Eighths in 2 wholes and 3 eighths ($16 + 3$)

Week 1 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • One Whole, Cut Up

OUT OF 11

Warm-Up • 2 | 4 | 8 | 2 | 4 | 1

Core • 3 | 6 | 1 | 1 | 3

Challenge • 4 | 2

1.2 • Same Number, New Name

OUT OF 11

Warm-Up • 3 | 2 | 4 | 2 | 6 | 2

Core • 9 | 8 | 3 | 2 | 4

Challenge • 2 | 9

1.3 • Which Is Bigger

OUT OF 11

Warm-Up • 2 | 3 | 3 | 3 | 6 | 7

Core • 10 | 6 | 15 | 8 | 9

Challenge • 21 | 5

1.4 • Adding Needs Matching Pieces

OUT OF 11

Warm-Up • 2 | 3 | 5 | 3 | 3 | 1

Core • 3 | 5 | 5 | 3 | 5

Challenge • 17 | 7

Fri • Fraction of a Set & Closest to One

ENRICHMENT

Challenge • 5 | 12 | 15 | 30 | 35 | 24 | 10 | 19

Same Number, New Clothes

Multiply top and bottom by the same thing and the value never moves. Type fractions like $\frac{3}{4}$.

Warm-Up 6 ITEMS

$\frac{1}{2} = \frac{?}{4}$ — the numerator

$\frac{1}{2} = \frac{?}{6}$

$\frac{1}{3} = \frac{?}{6}$

$\frac{1}{4} = \frac{?}{8}$

$\frac{2}{3} = \frac{?}{6}$

$\frac{3}{4} = \frac{?}{8}$

Core GRADED • SHOW YOUR WORKING

1. $\frac{2}{5} = \frac{?}{15}$

.....

2. $\frac{3}{8} = \frac{?}{24}$

.....

3. $\frac{5}{6} = \frac{?}{18}$

.....

4. $\frac{4}{7} = \frac{?}{21}$

.....

5. $\frac{7}{10} = \frac{?}{100}$

.....

Challenge NOT GRADED

C1. $\frac{9}{12}$ in simplest form — type as a/b

.....

C2. What was $\frac{1}{2}$ multiplied by to become $\frac{7}{14}$ — type the number

.....

Simplest Form

Divide top and bottom by the same thing until nothing else fits.

Warm-Up 6 ITEMS

2/4 in simplest form — type as a/b

3/6

4/8

2/6

5/10

3/9

Core GRADED • SHOW YOUR WORKING

1. 6/8

2. 10/15

3. 12/16

4. 18/24

5. 20/25

Challenge NOT GRADED

C1. 36/48

C2. The number you divide by to simplify 24/36 in one step

Spot the Equivalent

Cross-multiply, or find a common denominator. Either proof counts.

Warm-Up 6 ITEMS

Is $\frac{1}{2}$ the same as $\frac{2}{4}$ — yes or no

Is $\frac{1}{3}$ the same as $\frac{2}{6}$

Is $\frac{1}{2}$ the same as $\frac{1}{3}$

Is $\frac{3}{4}$ the same as $\frac{6}{8}$

Is $\frac{2}{5}$ the same as $\frac{4}{10}$

Is $\frac{2}{3}$ the same as $\frac{3}{4}$

Core GRADED • SHOW YOUR WORKING

1. Is $\frac{6}{9}$ the same as $\frac{8}{12}$ — yes or no

.....

2. Is $\frac{5}{8}$ the same as $\frac{10}{18}$

.....

3. Is $\frac{9}{12}$ the same as $\frac{15}{20}$

.....

4. $\frac{1}{2} \times \frac{5}{5}$ — type as a/b

.....

5. Is $\frac{7}{21}$ the same as $\frac{1}{3}$

.....

Challenge NOT GRADED

C1. Fractions equal to $\frac{1}{2}$ with denominators under 20 — how many

.....

C2. Is $\frac{12}{18}$ the same as $\frac{16}{24}$ — yes or no

.....

Fractions Equal to One

Any fraction with matching top and bottom is just 1 wearing a costume.

✦ Warm-Up 6 ITEMS

5/5 as a whole number

8/8

12/12

4/4

100/100

2/2

◆ Core GRADED • SHOW YOUR WORKING

1. $6/3$ as a whole number

2. $12/4$

3. $20/5$

4. How many quarters make 2

5. How many thirds make 3

★ Challenge NOT GRADED

C1. $7/2$ as a mixed number — the whole part

C2. $11/4$ — the whole part

Equivalence Wall

Build the strips and find every pair that lines up.

 Warm-Up 2 ITEMS

Halves in a whole

Quarters in a whole

 Core GRADED • SHOW YOUR WORKING

1. Quarters that line up with $\frac{1}{2}$ — type as a/b

.....
2. Sixths that line up with $\frac{1}{3}$

.....

 Challenge NOT GRADED

C1. Twelfths that line up with $\frac{3}{4}$ — type as a/b

.....

C2. Strips of 12 that line up with $\frac{2}{3}$

.....

C3. The smallest denominator where halves, thirds and quarters all line up

.....

Week 2 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Same Number, New Clothes

OUT OF 11

Warm-Up • 2 | 3 | 2 | 2 | 4 | 6

Core • 6 | 9 | 15 | 12 | 70

Challenge • $\frac{3}{4}$ | 7

2.2 • Simplest Form

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{3}$ Core • $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{4}{5}$ Challenge • $\frac{3}{4}$ | 12

2.3 • Spot the Equivalent

OUT OF 11

Warm-Up • yes | yes | no | yes | yes | no

Core • yes | no | yes | $\frac{5}{10}$ | yes

Challenge • 9 | yes

2.4 • Fractions Equal to One

OUT OF 11

Warm-Up • 1 | 1 | 1 | 1 | 1 | 1

Core • 2 | 3 | 4 | 8 | 9

Challenge • 3 | 2

Fri • Equivalence Wall

OUT OF 4

Warm-Up • 2 | 4

Core • $\frac{2}{4}$ | $\frac{2}{6}$ Challenge • $\frac{9}{12}$ | $\frac{8}{12}$ | 12

Common Denominators

Cut both into the same size pieces, then just compare the counts.

✦ Warm-Up 6 ITEMS

Common denominator of $\frac{1}{2}$ and $\frac{1}{4}$

Common denominator of $\frac{1}{3}$ and $\frac{1}{6}$

Common denominator of $\frac{1}{2}$ and $\frac{1}{3}$

$\frac{1}{2}$ as quarters — the numerator

$\frac{1}{3}$ as sixths — the numerator

Common denominator of $\frac{1}{4}$ and $\frac{1}{8}$

◆ Core GRADED • SHOW YOUR WORKING

1. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$

2. Common denominator of $\frac{3}{5}$ and $\frac{1}{2}$

3. $\frac{2}{3}$ as twelfths — the numerator

4. $\frac{3}{4}$ as twelfths — the numerator

5. Common denominator of $\frac{5}{6}$ and $\frac{3}{8}$

★ Challenge NOT GRADED

C1. Smallest common denominator of $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{1}{8}$

C2. Common denominator of $\frac{7}{10}$ and $\frac{2}{15}$

Benchmarks

Nearer 0, a half, or 1? Decide before you calculate anything.

Warm-Up 6 ITEMS

Is $\frac{1}{8}$ nearer 0 or 1 — type 0 or 1

Is $\frac{7}{8}$ nearer 0 or 1

$\frac{4}{8}$ simplified — type as a/b

Is $\frac{5}{6}$ nearer $\frac{1}{2}$ or 1 — type $\frac{1}{2}$ or 1

Is $\frac{1}{5}$ nearer 0 or $\frac{1}{2}$ — type 0 or $\frac{1}{2}$

Is $\frac{3}{5}$ more or less than $\frac{1}{2}$ — type more or less

Core GRADED • SHOW YOUR WORKING

1. Is $\frac{5}{9}$ more or less than $\frac{1}{2}$

.....

2. Is $\frac{4}{9}$ more or less than $\frac{1}{2}$

.....

3. Is $\frac{7}{16}$ more or less than $\frac{1}{2}$

.....

4. Is $\frac{9}{16}$ more or less than $\frac{1}{2}$

.....

5. Nearest benchmark to $\frac{11}{12}$ — type 0, $\frac{1}{2}$ or 1

.....

Challenge NOT GRADED

C1. Is $\frac{13}{25}$ more or less than $\frac{1}{2}$

.....

C2. Nearest benchmark to $\frac{8}{15}$ — type 0, $\frac{1}{2}$ or 1

.....

Comparing & Ordering

Common denominator, or reason from benchmarks. Say which you used.

✦ Warm-Up 6 ITEMS

Larger: $\frac{1}{2}$ or $\frac{1}{3}$ — type it

Larger: $\frac{3}{4}$ or $\frac{1}{2}$

Larger: $\frac{1}{4}$ or $\frac{1}{8}$

Larger: $\frac{2}{3}$ or $\frac{1}{3}$

Smaller: $\frac{1}{5}$ or $\frac{1}{6}$

Larger: $\frac{5}{8}$ or $\frac{3}{8}$

◆ Core GRADED • SHOW YOUR WORKING

1. Larger: $\frac{2}{3}$ or $\frac{3}{4}$ — type it

.....

2. Larger: $\frac{3}{5}$ or $\frac{5}{8}$

.....

3. Larger: $\frac{5}{6}$ or $\frac{7}{9}$

.....

4. Smallest of $\frac{1}{3}$, $\frac{2}{5}$, $\frac{1}{4}$ — type it

.....

5. Largest of $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{12}$

.....

★ Challenge NOT GRADED

C1. Order $\frac{3}{4}$, $\frac{5}{6}$, $\frac{7}{8}$ — type the largest

.....

C2. Larger: $\frac{11}{15}$ or $\frac{7}{10}$ — type it

.....

On the Number Line

Every fraction has one address between 0 and 1. Find it.

✦ Warm-Up 6 ITEMS

Halfway between 0 and 1 — type as a/b

Halfway between 0 and $\frac{1}{2}$

Halfway between $\frac{1}{2}$ and 1

Marks between 0 and 1 if split into quarters

Marks if split into fifths

$\frac{1}{2}$ as a decimal

◆ Core GRADED • SHOW YOUR WORKING

1. $\frac{3}{4}$ as a decimal

.....

2. $\frac{1}{4}$ as a decimal

.....

3. $\frac{1}{5}$ as a decimal

.....

4. $\frac{3}{5}$ as a decimal

.....

5. Halfway between $\frac{1}{4}$ and $\frac{1}{2}$ — type as a/b

.....

★ Challenge NOT GRADED

C1. $\frac{7}{8}$ as a decimal

.....

C2. A fraction exactly between $\frac{1}{3}$ and $\frac{2}{3}$ — type as a/b

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 2 ITEMS

$1/2 = ?/8$ — the numerator

$4/8$ simplified — type as a/b

◆ Core GRADED • SHOW YOUR WORKING

1. $6/8$ simplified

.....

2. Common denominator of $2/3$ and $3/4$

.....

3. Larger: $3/5$ or $5/8$ — type it

.....

4. Is $5/9$ more or less than $1/2$

.....

5. $3/4$ as a decimal

.....

★ Challenge NOT GRADED

C1. Order $3/4$, $5/6$, $7/8$ — type the largest

.....

Week 3 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Common Denominators

OUT OF 11

Warm-Up • 4 | 6 | 6 | 2 | 2 | 8

Core • 12 | 10 | 8 | 9 | 24

Challenge • 24 | 30

3.2 • Benchmarks

OUT OF 11

Warm-Up • 0 | 1 | $\frac{1}{2}$ | 1 | 0 | more

Core • more | less | less | more | 1

Challenge • more | $\frac{1}{2}$

3.3 • Comparing & Ordering

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{1}{6}$ | $\frac{5}{8}$

Core • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{5}{6}$ | $\frac{1}{4}$ | $\frac{2}{3}$

Challenge • $\frac{7}{8}$ | $\frac{11}{15}$

3.4 • On the Number Line

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 3 | 4 | 0.5

Core • 0.75 | 0.25 | 0.2 | 0.6 | $\frac{3}{8}$

Challenge • 0.875 | $\frac{1}{2}$

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 4 | $\frac{1}{2}$

Core • $\frac{3}{4}$ | 12 | $\frac{5}{8}$ | more | 0.75

Challenge • $\frac{7}{8}$

Adding Like Denominators

Same size pieces already. Add the counts, leave the bottom alone.

Warm-Up 6 ITEMS

$1/4 + 1/4$ — type as a/b

$1/5 + 2/5$

$1/8 + 3/8$

$2/6 + 1/6$

$1/3 + 1/3$

$3/10 + 2/10$

Core GRADED • SHOW YOUR WORKING

1. $3/8 + 3/8$ _____

2. $5/12 + 3/12$ _____

3. $7/10 + 1/10$ _____

4. $2/9 + 4/9$ _____

5. $5/6 + 1/6$ _____

Challenge NOT GRADED

C1. $7/8 + 5/8$ — type as a/b

C2. $3/5 + 4/5$

Subtracting Like Denominators

Same move, other direction.

✦ Warm-Up 6 ITEMS

$3/4 - 1/4$ — type as a/b

$4/5 - 1/5$

$7/8 - 3/8$

$5/6 - 1/6$

$2/3 - 1/3$

$9/10 - 4/10$

◆ Core GRADED • SHOW YOUR WORKING

1. $1 - 1/4$

2. $1 - 3/8$

3. $11/12 - 5/12$

4. $1 - 2/5$

5. $7/9 - 1/9$

★ Challenge NOT GRADED

C1. $2 - 3/4$ — type as a/b
.....

C2. $1\frac{1}{2} - 3/4$ — type as a/b
.....

Unlike Denominators

The grade-five ceiling. Convert first, then add.

✦ Warm-Up 6 ITEMS

$1/2 + 1/4$ — type as a/b

$1/3 + 1/6$

$1/2 + 1/8$

$1/4 + 1/8$

$1/2 - 1/4$

$2/3 - 1/6$

◆ Core GRADED • SHOW YOUR WORKING

1. $1/2 + 1/3$

2. $3/4 + 1/8$

3. $2/3 + 1/4$

4. $3/4 - 1/3$

5. $5/6 - 1/2$

★ Challenge NOT GRADED

C1. $2/3 + 3/4$ — type as a/b

.....

C2. $1/2 + 1/3 + 1/6$

.....

Fractions in Words

Read it twice, then decide whether it is an add or a subtract.

Warm-Up 6 ITEMS

Ate $\frac{1}{4}$ of a pizza, then $\frac{1}{4}$ more — total, as a/b

Had $\frac{3}{4}$, ate $\frac{1}{4}$ — left

$\frac{1}{2}$ hour plus $\frac{1}{2}$ hour — hours

$\frac{2}{5}$ of a jug plus $\frac{1}{5}$ — as a/b

1 whole minus $\frac{1}{3}$ — as a/b

$\frac{1}{8} + \frac{1}{8}$ — as a/b

Core GRADED • SHOW YOUR WORKING

1. Walked $\frac{1}{2}$ mile then $\frac{1}{4}$ mile — total, as a/b

2. A $\frac{3}{4}$ cup recipe, you have $\frac{1}{2}$ — how much more, as a/b

3. Read $\frac{2}{3}$ of a book, then $\frac{1}{6}$ more — total, as a/b

4. Ran $\frac{5}{6}$ km and walked $\frac{1}{3}$ km — total, as a/b

5. 1 hour minus $\frac{1}{4}$ hour — minutes

Challenge NOT GRADED

C1. Three pieces of $\frac{3}{8}$ — total, as a/b

C2. $\frac{2}{3}$ of an hour in minutes

Closest to One

Draw cards, build a fraction, get nearest to 1 without going over.

★ Warm-Up 2 ITEMS

Which is nearer 1: $\frac{3}{4}$ or $\frac{1}{2}$ — type it

Which is nearer 1: $\frac{7}{8}$ or $\frac{3}{4}$

◆ Core GRADED • SHOW YOUR WORKING

1. From digits 5 and 6, the fraction under 1 — type as a/b

.....

2. From 7 and 8, the fraction under 1

.....

★ Challenge NOT GRADED

C1. Nearer 1: $\frac{8}{9}$ or $\frac{9}{10}$ — type it

.....
C2. How far is $\frac{9}{10}$ from 1 — type as a/b

.....
C3. From digits 3, 4, 5, 6 the closest to 1 without going over — type as a/b

.....

Week 4 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Adding Like Denominators

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$

Core • $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{4}{5}$ | $\frac{2}{3}$ | 1

Challenge • $\frac{3}{2}$ | $\frac{7}{5}$

4.2 • Subtracting Like Denominators

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{1}{2}$

Core • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{2}{3}$

Challenge • $\frac{5}{4}$ | $\frac{3}{4}$

4.3 • Unlike Denominators

OUT OF 11

Warm-Up • $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{3}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$

Core • $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{11}{12}$ | $\frac{5}{12}$ | $\frac{1}{3}$

Challenge • $\frac{17}{12}$ | 1

4.4 • Fractions in Words

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | $\frac{3}{5}$ | $\frac{2}{3}$ | $\frac{1}{4}$

Core • $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{5}{6}$ | $\frac{7}{6}$ | 45

Challenge • $\frac{9}{8}$ | 40

Fri • Closest to One

OUT OF 4

Warm-Up • $\frac{3}{4}$ | $\frac{7}{8}$

Core • $\frac{5}{6}$ | $\frac{7}{8}$

Challenge • $\frac{9}{10}$ | $\frac{1}{10}$ | $\frac{5}{6}$

Fraction of a Set

Divide by the bottom, multiply by the top.

✦ Warm-Up 6 ITEMS

1/2 of 10

1/3 of 12

1/4 of 20

1/5 of 25

1/2 of 30

1/6 of 30

◆ Core GRADED • SHOW YOUR WORKING

1. $\frac{2}{5}$ of 30

.....

2. $\frac{3}{4}$ of 20

.....

3. $\frac{2}{3}$ of 18

.....

4. $\frac{5}{6}$ of 24

.....

5. $\frac{3}{8}$ of 40

.....

★ Challenge NOT GRADED

C1. $\frac{2}{3}$ of 45

.....

C2. $\frac{7}{10}$ of 120

.....

Of Money and Time

The same operator, applied to dollars and minutes.

✦ Warm-Up 6 ITEMS

1/2 of \$20

1/4 of \$40

1/2 of an hour in minutes

1/4 of an hour in minutes

1/3 of \$30

1/5 of \$50

◆ Core GRADED • SHOW YOUR WORKING

1. $\frac{3}{4}$ of \$60

2. $\frac{2}{3}$ of an hour in minutes

3. $\frac{3}{5}$ of \$45

4. $\frac{5}{6}$ of an hour in minutes

5. $\frac{2}{5}$ of \$75

★ Challenge NOT GRADED

C1. $\frac{3}{8}$ of \$96

C2. $\frac{7}{12}$ of an hour in minutes

Working Backwards

Given the part, find the whole.

Warm-Up 6 ITEMS

Half a number is 6. The number

A third is 4. The number

A quarter is 5. The number

A fifth is 3. The number

Half is 15. The number

A sixth is 2. The number

Core GRADED • SHOW YOUR WORKING

1. $\frac{2}{3}$ of a number is 12. The number

.....

2. $\frac{3}{4}$ of a number is 15. The number

.....

3. $\frac{2}{5}$ of a number is 10. The number

.....

4. $\frac{5}{6}$ of a number is 20. The number

.....

5. $\frac{3}{8}$ of a number is 9. The number

.....

Challenge NOT GRADED

C1. $\frac{2}{7}$ of a number is 18. The number

.....

C2. After spending $\frac{3}{5}$, \$20 is left. The starting amount

.....

Mixed Fraction Problems

No signposting. Read, decide, then compute.

Warm-Up 6 ITEMS

1/2 of 16

$\frac{1}{4} + \frac{1}{4}$ — type as a/b

$1 - \frac{1}{2}$ — as a/b

$\frac{2}{6}$ simplified — as a/b

$\frac{1}{3}$ of 15

$\frac{3}{4} - \frac{1}{4}$ — as a/b

Core GRADED • SHOW YOUR WORKING

1. $\frac{2}{3} + \frac{1}{6}$ — as a/b

2. $\frac{3}{4}$ of 24

3. $\frac{5}{8} - \frac{1}{4}$ — as a/b

4. $\frac{2}{5}$ of a number is 8. The number

5. Larger: $\frac{5}{8}$ or $\frac{2}{3}$ — type it

Challenge NOT GRADED

C1. $\frac{2}{3} + \frac{3}{4}$ — as a/b

C2. $\frac{7}{12}$ of 60

Fold the Universe, Built

Finish the paper number line with every fraction landed on it.

★ Warm-Up 2 ITEMS

Folds to get halves

Folds to get quarters

◆ Core GRADED • SHOW YOUR WORKING

1. Folds to get eighths

2. Sections after 4 folds

★ Challenge NOT GRADED

C1. Sections after 6 folds

C2. Folds needed for 32 sections

C3. Marks strictly between 0 and 1 on a sixteenths line

Week 5 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 • Fraction of a Set

OUT OF 11

Warm-Up • 5 | 4 | 5 | 5 | 15 | 5

Core • 12 | 15 | 12 | 20 | 15

Challenge • 30 | 84

5.2 • Of Money and Time

OUT OF 11

Warm-Up • 10 | 10 | 30 | 15 | 10 | 10

Core • 45 | 40 | 27 | 50 | 30

Challenge • 36 | 35

5.3 • Working Backwards

OUT OF 11

Warm-Up • 12 | 12 | 20 | 15 | 30 | 12

Core • 18 | 20 | 25 | 24 | 24

Challenge • 63 | 50

5.4 • Mixed Fraction Problems

OUT OF 11

Warm-Up • 8 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | 5 | $\frac{1}{2}$ Core • $\frac{5}{6}$ | 18 | $\frac{3}{8}$ | 20 | $\frac{2}{3}$ Challenge • $\frac{17}{12}$ | 35

Fri • Fold the Universe, Built

OUT OF 4

Warm-Up • 1 | 2

Core • 3 | 16

Challenge • 64 | 5 | 15

The Equivalence Proof

Show the same amount two different ways and say why they match.

Warm-Up 6 ITEMS

$1/2 = ?/4$ — the numerator

$2/4$ simplified — type as a/b

$1/2 \times 3/3$ — type as a/b

$3/6$ simplified

$3/3$ as a whole number

Multiplying by 1 changes a number by how much

Core GRADED • SHOW YOUR WORKING

1. $2/3 \times 4/4$ — type as a/b

2. $8/12$ simplified

3. $5/6 \times 3/3$

4. $15/18$ simplified

5. What $3/4$ was multiplied by to become $12/16$

Challenge NOT GRADED

C1. $18/27$ simplified — type as a/b

C2. Two fractions equal to $4/6$ — type the simplest

Number Line Defence

Every fraction has one place. Put it there and argue for it.

Warm-Up 6 ITEMS

1/2 as a decimal

1/4 as a decimal

3/4 as a decimal

1/5 as a decimal

2/5 as a decimal

1/10 as a decimal

Core GRADED • SHOW YOUR WORKING

1. $\frac{3}{8}$ as a decimal

.....

2. $\frac{5}{8}$ as a decimal

.....

3. $\frac{7}{10}$ as a decimal

.....

4. Which sits further right: $\frac{2}{3}$ or 0.6 — type it

.....

5. $\frac{3}{5}$ as a decimal

.....

Challenge NOT GRADED

C1. $\frac{7}{8}$ as a decimal

.....

C2. Between 0.3 and 0.4 — type $\frac{1}{3}$ or $\frac{2}{5}$, whichever fits

.....

Mission Review

Everything from six weeks, mixed together.

Warm-Up 6 ITEMS

6/8 simplified — type as a/b

$\frac{1}{4} + \frac{1}{4}$ — as a/b

$\frac{1}{2}$ of 14

$1 - \frac{1}{3}$ — as a/b

Larger: $\frac{1}{2}$ or $\frac{1}{3}$ — type it

$\frac{1}{2}$ as a decimal

Core GRADED • SHOW YOUR WORKING

1. $\frac{2}{3} + \frac{1}{4}$ — as a/b

2. $\frac{3}{4}$ of 32

3. Common denominator of $\frac{5}{6}$ and $\frac{3}{8}$

4. $\frac{3}{4} - \frac{1}{3}$ — as a/b

5. $\frac{2}{5}$ of a number is 10. The number

Challenge NOT GRADED

C1. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$ — as a/b

C2. $\frac{7}{12}$ of 60

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

2/4 simplified — type as a/b

$\frac{1}{3} + \frac{1}{3}$ — as a/b

$\frac{1}{4}$ of 16

Is $\frac{3}{6}$ the same as $\frac{1}{2}$ — yes or no

$\frac{1}{5}$ as a decimal

$\frac{5}{6} - \frac{1}{6}$ — as a/b

◆ Core GRADED • SHOW YOUR WORKING

1. $\frac{1}{2} + \frac{1}{3}$ — as a/b

2. $\frac{12}{16}$ simplified

3. $\frac{5}{6}$ of 24

4. Larger: $\frac{2}{3}$ or $\frac{3}{4}$ — type it

5. $\frac{3}{8}$ as a decimal

★ Challenge NOT GRADED

C1. $\frac{2}{3} + \frac{3}{4}$ — as a/b

C2. After spending $\frac{3}{5}$, \$20 is left. The starting amount

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

✦ Warm-Up 2 ITEMS

4/8 simplified — type as a/b

$\frac{1}{4} + \frac{1}{4}$ — as a/b

◆ Core GRADED • SHOW YOUR WORKING

1. $\frac{1}{2} = ?/10$ — the numerator

2. 10/15 simplified

3. $\frac{1}{2} + \frac{1}{4}$ — as a/b

4. $\frac{3}{4} - \frac{1}{3}$ — as a/b

5. Larger: $\frac{3}{5}$ or $\frac{5}{8}$ — type it

6. $\frac{2}{3}$ of 18

7. $\frac{3}{4}$ as a decimal

8. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$

★ Challenge NOT GRADED

C1. $\frac{2}{3} + \frac{3}{4}$ — as a/b

C2. $\frac{3}{8}$ of a number is 9. The number

Week 6 Answers

Mission 04 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

6.1 • The Equivalence Proof

OUT OF 11

Warm-Up • 2 | $\frac{1}{2}$ | $\frac{3}{6}$ | $\frac{1}{2}$ | 1 | 0

Core • $\frac{8}{12}$ | $\frac{2}{3}$ | $\frac{15}{18}$ | $\frac{5}{6}$ | 4

Challenge • $\frac{2}{3}$ | $\frac{2}{3}$

6.2 • Number Line Defence

OUT OF 11

Warm-Up • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1

Core • 0.375 | 0.625 | 0.7 | $\frac{2}{3}$ | 0.6

Challenge • 0.875 | $\frac{1}{3}$

6.3 • Mission Review

OUT OF 11

Warm-Up • $\frac{3}{4}$ | $\frac{1}{2}$ | 7 | $\frac{2}{3}$ | $\frac{1}{2}$ | 0.5

Core • $\frac{11}{12}$ | 24 | 24 | $\frac{5}{12}$ | 25

Challenge • 1 | 35

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • $\frac{1}{2}$ | $\frac{2}{3}$ | 4 | yes | 0.2 | $\frac{2}{3}$

Core • $\frac{5}{6}$ | $\frac{3}{4}$ | 20 | $\frac{3}{4}$ | 0.375

Challenge • $\frac{17}{12}$ | 50

Fri • Mission 04 Test

OUT OF 10

Warm-Up • $\frac{1}{2}$ | $\frac{1}{2}$

Core • 5 | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{5}{12}$ | $\frac{5}{8}$ | 12 | 0.75 | 12

Challenge • $\frac{17}{12}$ | 24